Cristhian Núñez Ramos

Curriculum Vitae

1. Education

2022-2025 **Ph.D. in Mathematics**, *Pontificia Universidad Católica de Chile*, Santiago-Chile.
Title of Thesis: Symplectic Hamiltonian Discontinuous Galerkin Methods for Shallow Water Equations. Advisor: Manuel A. Sánchez

2021–2022 **Master in Mathematics**, *Pontificia Universidad Católica de Chile*, Santiago-Chile.

2015–2018 **Master in Optimization**, *Escuela Politécnica Nacional*, Quito-Ecuador.
Title of Thesis: Numerical Simulation of Free Boundary Bingham Fluids. Advisor: Pedro Merino

2009–2014 **Bachelor in Pure Mathematics**, *Escuela Politécnica Nacional*, Quito-Ecuador.
Title of Thesis: A Proper Orthogonal Decomposition for the Fisher Equation. Advisor: Pedro Merino

2. Areas of interest

My research focuses on Numerical Analysis of Partial Differential Equations, particularly Finite Element Methods. Specifically, I am interested in Discontinuous Galerkin Methods, Hybridizable Discontinuous Galerkin Methods, and the development of novel numerical methods for Hamiltonian systems that preserve key physical invariants in long-term simulations. Additionally, my interests extend to the optimal control of PDEs, with applications in mathematical modeling, computational physics, and Engineering problems requiring accurate and robust simulations.

3. Research Experience

2024 **Researcher (Internship)**, *School of Mathematics, University of Minnesota*, Minneapolis, United States.
Developed numerical methods for solving the vector Laplacian equation.

2021–2025 **Ph.D. Student**, *Facultad de Matemáticas, Pontificia Universidad Católica de Chile*, Santiago, Chile.
Developed novel numerical methods for shallow water equations.

2015–2021 **Researcher**, *Centro de Modelización Matemática, Escuela Politécnica Nacional*, Quito, Ecuador.
Developed and analyzed mathematical models and algorithms for viscoplastic fluid simulations.

2015–2016 **Researcher**, *Facultad de Ciencias – Instituto Geofísico del Ecuador*, Quito, Ecuador.
Designed algorithms for image classification and modeling of volcanic lava and viscoplastic flows.

4. Academic Appointments

25/08/2025 **Postdoctoral Associate**, *University of Minnesota*, Minneapolis, USA.
Menthor: Bernardo Cockburn.
We are developing post-processing techniques for elliptic problems that systematically enhance the quality of approximations produced by Galerkin methods by exploiting hidden properties of the governing equations.

5. Published Papers (Refereed Journals)

1. Merino, P. M., & Núñez, C. A. (2024). Numerical simulation of free boundary biviscous fluids. Bulletin of Computational Applied Mathematics, 12(2), 163–188.
2. Núñez, C., & Sánchez, M. A. (2025). Symplectic Hamiltonian hybridizable discontinuous Galerkin methods for linearized shallow water equations. Computer Methods in Applied Mechanics and Engineering, 447, 118383. https://doi.org/10.1016/j.cma.2025.118383

6. Manuscripts Submitted

1. *HDG methods for the two-dimensional vector Laplacian* with Bernardo Cockburn and Manuel Sánchez.

7. Manuscripts in Preparation

1. *Turbo Post-Processing for Mixed Methods for One-dimensional Elliptic Problems* with Bernardo Cockburn.
2. *Symplectic Hamilton–HDG Methods for Nonlinear Shallow Water Equations* with Manuel Sánchez.
3. *POD Techniques for an Optimal Control Problem problem of Fisher equation* with Pedro Merino.

8. Other Publications

1. Núñez Ramos, C. A., & Villacís Tulcán, N. E. (2025). Simulation of an optimal control of the epidemiological reservoir–people model. Matemática (ESPOL–FCNM Journal), 23(2).
2. Navarrete Vaca, L., & Núñez Ramos, Cristhian (2025). Control and communication in dialogue: controllability, observability, and bang–bang dynamics. Selecciones Matemáticas, 12(2), 365–374. https://doi.org/10.17268/sel.mat.2025.02.08
3. Núñez Ramos, C. A., & Pizanan Cárdenas, C. H. (2026). Optimal control of the Fisher equation: a comparison of steepest descent and quasi-Newton methods. Matemática (ESPOL–FCNM Journal), 24(1).

9. Presentations

August 11–15, 2025 **XII International Congress on Applied and Computational Mathematics (XII CIMAC)**, *Universidad Nacional del Altiplano*, Puno, Perú.
Oral presentation: "HDG Methods for the two-dimensional Vector Laplacian Equation"

March 25–28, 2025 **1st Heidelberg–Chile Workshop on Scientific Computing**, Santiago, Chile.
Poster presentation: "Hybridizable Discontinuous Galerkin Methods for the Vector Laplacian Equation"

March 17–20, 2025 **IACM Computational Fluids Conference (CFC)**, Santiago, Chile.
Oral presentation: "Symplectic Hamiltonian HDG Methods for Linearized Shallow Water Equations"

June 12–14, 2024 **Conferencias Colombianas de Matemáticas Aplicadas e Industriales (MAPI3)**, Bucaramanga, Colombia.
Oral presentation: "Symplectic Hamiltonian HDG Methods for Linearized Shallow Water Equations"

April 24–26, 2024 **XXXVI Jornada de Matemática de la Zona Sur**, *Universidad Católica de Temuco*, Temuco, Chile.
Oral presentation: "Symplectic Hamiltonian HDG Methods for Shallow Water Equations"

January 15–19, 2024 **Workshop on Numerical Analysis of Partial Differential Equations (WONAPDE)**, *Universidad de Concepción*, Concepción, Chile.
Oral presentation: "Symplectic Hamiltonian HDG Methods for Linearized Shallow Water Equations"

October 10–13, 2023 **XLIX Semana de la Matemática**, *Pontificia Universidad Católica de Valparaíso*, Valparaíso, Chile.
Oral presentation: "Métodos Hamiltonianos Simplécticos HDG para las ecuaciones de aguas poco profundas"

May 29–31, 2023 **Encuentro Latinoamericano de Estudiantes de Doctorado en Modelamiento, Ingeniería y Ciencias (EMIC)**, *Universidad de Chile*, Santiago, Chile.
Oral presentation: "Symplectic Hamiltonian HDG Methods for Shallow Water Equations"

November, 14–19, 2022 **XVII Encuentro de Matemática y sus Aplicaciones**, *Escuela Politécnica Nacional*, Quito, Ecuador.
Oral presentation: "Numerical Simulation of Free Boundary Biviscous Fluids"

September 3-7, 2018 **VI Latin American Workshop on Optimization and Control**, *Escuela Politécnica Nacional*, Quito, Ecuador.
Oral presentation: "Numerical Simulation of Free Boundary Bingham Fluids"

November, 11th 2015 **III Concurso de Reconocimiento a la Investigación Universitaria Estudiantil – Galardones Nacionales**, Quito, Ecuador.
Oral presentation: "Reducción de un Problema de Control Óptimo usando técnicas POD"

10. Academic Service

2025 Workshop on Applied Mathematics, Pontificia Universidad Católica de Chile, Santiago, Chile. Organizer. Responsible for scientific coordination, invited speakers, program design, and workshop logistics.`https://appliedmathuc.github.io/workshop-applied-math-2025/`

11. Workshops attended

2023 XXXV Jornada de Matemática de la Zona Sur, Universidad de Concepción, Chile

2023 2nd Chile-Japan Workshop on Mathematical Physics and PDE, Universidad de Santiago de Chile.

2022 Workshop on Scientific Machine Learning: Mathematics and Applications, Santiago, Chile.

2022 Workshop on Minimum Residual and Least-Squares Finite Element Methods, Santiago, Chile.

12. Teaching Experience

10/01/2018–04/05/2021 **Adjunct Professor**, *Escuela Politécnica Nacional*, Quito, Ecuador.
Courses taught:
- Numerical Analysis for PDEs (2018-II, two sections)
- Numerical Analysis I (2018-II, two sections)
- Numerical Analysis II (2018-II, two sections)
- Linear Algebra (2018- I-II, two sections)
- Calculus of variations (2019- I-II, two sections)
- Modelling of PDEs (2019- I-II, two sections)
- Numerical Analysis II (2019-I-II, two sections)
- Biomathematics (2020-I-II, one sections)
- Numerical Analysis for PDEs (2020-I-II, one section)
- Non-Linear Analysis (2021-I, two sections)
- Numerical Analysis for PDEs (2021-I, one section)
- Biomathematics (2021-I, one section)

10/16/2017-02/28/2019 **Adjunct Professor**, *Full Time*, Escuela Politécnica Nacional, Quito-Ecuador.
Courses taught:
- Numerical Analysis I (2017-II, two sections)
- Numerical Analysis II (2017-II, two sections)
- Algebra (2017-II, one section)
- Calculus II (2018-I, two sections)
- Optimization I (2018-I, two sections)
- Differential Equations (2018-I, two sections)
- Calculus II (2018-II, two sections)
- Optimization I (2018-II, two sections)
- Optimization I (2018-II, two sections)

01/03/2017-09/11/2017 **Adjunct Professor**, *Full Time*, Escuela Politécnica Nacional, Quito-Ecuador.
Courses taught:
- Numerical Analysis II (2017-I, two sections)
- Calculus I (2017-II, two sections)
- Calculus II (2017-I, two sections)
- Complex Calculus (2017-I, four sections)
- Analysis I (2017-I, two sections)

03/05/2025–07/14/2025 **Teaching Assistant**, *Pontificia Universidad Católica de Chile*, Santiago, Chile.
Teaching assistance during doctoral studies, including support in lectures and grading in the course of Fundamentos Matemáticos para la Inteligencia Artificial.

09/01/2009–04/30/2012 **High School Teacher**, *Plan Internacional Programa Bolívar*, Guaranda, Ecuador.
Teaching Mathematics and Geometry at secondary education level.

13. Professional and Consulting Experience

2017–2018 **Sampling Analyst (Consulting Services)**, *Instituto Nacional de Evaluación Educativa*, Quito, Ecuador.
Development of data-driven methods to support educational evaluation and the design of analytical tools for measuring academic progress.

09/01/2014–03/31/2015 **Sampling Assistant**, *Instituto Nacional de Estadísticas y Censos*, Quito, Ecuador.
Design and implementation of statistical sampling methodologies for unemployment studies in Ecuador.

14. Early Career Experience

02/03/2014– 06/30/2014 **Undergraduate Intern**, *Escuela Politécnica Nacional*, Quito, Ecuador.
Teaching assistance in Mathematics and Geometry and academic support for undergraduate students.

14. Memberships and Awards

- Postdoctoral Associate. School of Mathematics, University of Minnesota (2025 –)
- VRI–UC Research Scholarship, Pontificia Universidad Católica de Chile (2025).
- Gubernamental ANID fellowship for doctoral studies in Mathematics. Chile (2021-2024)
- Associate Researcher of Centro de Modelización Matemática - Escuela Politécnica Nacional - Ecuador (2015-2021)
- Member of Sociedad Ecuatoriana de Matemáticas - Ecuador (2015–)
- PIGR-22-03 project grant, awarded by the Vicerectorado de Investigación, Escuela Politécnica Nacional, for research in Mathematical Optimization. (2016-2018)
- EPN fellowship for master studies in Optimization. Escuela Politénica Nacional. Ecuador (2015-2017)

15. Skills

Comunication **Languages**, *Spanish: Native*, English: Professional working proficiency.

Programming Languages **Technical Skills**, Python, Matlab, SPSS, Stata, R/RStudio, C/C++, SQL, SML, Scheme, SQL, LateX.